

Week 5

Other project evaluation techniques

This Week We will learn:

Other evaluation technique: Internal Rate of Return (IRR) - SLO1 & SLO4

Other evaluation technique: Accounting Rate of Return (ARR) - SLO1 & SLO4

Other evaluation technique: Payback Period - SLO1 & SLO4

Multiple Project evaluation technique: Capital rationing and Profitability index (PI) - SLO1 & SLO4

Learning Objectives:

At the end of this module you will:

SLO1: describe the principles and objectives of financial decision-making

SLO4: evaluate a corporate investment proposal and provide a recommendation to accept or reject it

Net Present Value & Its Rules

- Net Present Value (NPV) =
Total PV of future project CF's less the Initial Investment
- Estimating NPV:
 1. Estimate future cash flows: how much? and when?
 2. Estimate discount rate
 3. Estimate initial costs
- Minimum Acceptance Criteria: Accept if $NPV > 0$
- Ranking Criteria: Choose the highest NPV

Net Present Value: Example

- Suppose Big Deal Co. has an opportunity to make an investment of \$100,000 that will return \$33,000 in year 1, \$38,000 in year 2, \$43,000 in year 3, \$48,000 in year 4, and \$53,000 in year 5. If the company's required return is 12% should they make the investment?

	Cash	PV of
Year	Flow	Cash Flow
0	\$ (100,000)	\$ (100,000)
1	\$ 33,000	\$ 29,464
2	\$ 38,000	\$ 30,293
3	\$ 43,000	\$ 30,607
4	\$ 48,000	\$ 30,505
5	\$ 53,000	\$ 30,074
Net Present Value		\$ 50,943

Answer: YES! The NPV is greater than \$0. Therefore, the investment does return at least the required rate of return.

Why Use Net Present Value?

- Accepting positive NPV projects benefits shareholders.
 - ✓ NPV uses cash flows
 - ✓ NPV uses all relevant cash flows of the project
 - ✓ NPV discounts the cash flows properly
- Reinvestment assumption: the NPV rule assumes that all cash flows can be reinvested at the discount rate.

Calculating NPV with Spreadsheets

- Spreadsheets are an excellent way to compute NPVs, especially when you have to compute the cash flows as well.
- Using the NPV function:
 - The first component is the required return entered as a decimal.
 - The second component is the range of cash flows *beginning with year 1*.
 - Add the initial investment after computing the NPV.

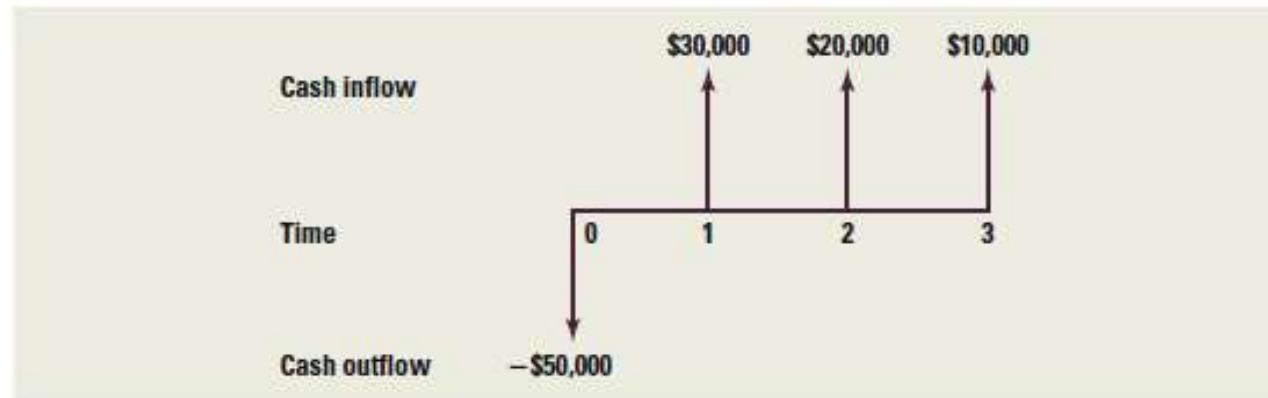


The Payback Period Method

- How long does it take the project to “pay back” its initial investment?
- Payback Period = number of years to recover initial costs
- Minimum Acceptance Criteria:
 - Set by management; a predetermined time period
- Ranking Criteria:
 - Set by management; often the shortest payback period is preferred

Example: Payback Method

- Consider a project with an investment of \$50,000 and cash inflows in years 1,2, & 3 of \$30,000, \$20,000, \$10,000



- The timeline above clearly illustrates that payback in this situation is 2 years. The first two years of return = \$50,000 which exactly “pays back” the initial investment

Example: Payback Method

• Year	A	B	C
0	-100	-100	-100
1	20	50	50
2	30	30	30
3	50	20	20
4	60	60	60,000
Payback			
Period	3	3	3

The Payback Period Method

- Disadvantages:
 - It fails to account for the time value of money
 - The threshold payback period for accepting projects is arbitrary
 - It is biased against projects that produce their largest cash flows late in their lives
 - It ignores cash flows that occur after payback has been achieved
- Advantages:
 - It is easy to calculate, although it requires cash flow forecasts
 - It is easy to understand
 - It favours projects that provide short-term liquidity
 - It provides some insight into risk: Cash flows received in the distant future are inherently more risky (i.e. they are less certain). Hence, a shorter payback period is associated with lower risk
 - It can provide useful supplementary information
 - It can be used to evaluate smaller projects

The Discounted Payback Period

- How long does it take the project to “pay back” its initial investment, taking the time value of money into account?
- Decision rule: Accept the project if it pays back on a discounted basis within the specified time.
- By the time you have discounted the cash flows, you might as well calculate the NPV.

Example: Discounted Payback Period

- Suppose Big Deal Co. has an opportunity to make an investment of \$100,000 that will return \$33,000 in year 1, \$38,000 in year 2, \$43,000 in year 3, \$48,000 in year 4, and \$53,000 in year 5. If the company's required return is 12% and predetermined payback period is 3 years should they make the investment?

	Cash		PV of		Cumulative
Year	Flow		Cash Flow		PV of Cash Flows
0	\$	(100,000)	\$	(100,000)	\$ (100,000)
1	\$	33,000	\$	29,464	\$ (70,536)
2	\$	38,000	\$	30,293	\$ (40,242)
3	\$	43,000	\$	30,607	\$ (9,636)
4	\$	48,000	\$	30,505	\$ 20,869
5	\$	53,000	\$	30,074	\$ 50,943

Answer: **NO!** At the end of three years the project has still not broken even or “paid back”. Therefore, it must be rejected.

Average Rate of Return (ARR)

$$\text{ARR} = \frac{\text{Average Net Income}}{\text{Average Book Value of Investment}}$$

- STEP 1: Determining average net income
- STEP 2: Determining average investment
- STEP 3: Determining ARR

Accounting Rate of Return

- Another attractive, but fatally flawed, approach
- Ranking Criteria and Minimum Acceptance Criteria set by management

Example: Accounting Rate of Return

Consider a company that is evaluating whether to buy a store in a new mall. The purchase price is \$500,000. We will assume that the store has an estimated life of five years and will need to be completely scrapped or rebuilt at the end of that time. The projected yearly sales and expense figures are shown in Table 7.2.

TABLE 7.2

Projected Yearly Revenue and Costs for Average Accounting Return

	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5
Revenue	\$433,333	\$450,000	\$266,667	\$200,000	\$133,333
Expenses	<u>200,000</u>	<u>150,000</u>	<u>100,000</u>	<u>100,000</u>	<u>100,000</u>
Before-tax cash flow	233,333	300,000	166,667	100,000	33,333
Depreciation	<u>100,000</u>	<u>100,000</u>	<u>100,000</u>	<u>100,000</u>	<u>100,000</u>
Earnings before taxes	133,333	200,000	66,667	0	— 66,667
Taxes ($t_c = .25$)*	<u>33,333</u>	<u>50,000</u>	<u>16,667</u>	<u>0</u>	<u>— 16,667</u>
Net income	\$100,000	\$150,000	\$ 50,000	\$ 0	—\$ 50,000

$$\text{Average net income} = \frac{(\$100,000 + 150,000 + 50,000 + 0 - 50,000)}{5} = \$50,000$$

$$\text{Average investment} = \frac{(\$500,000 + 400,000 + 300,000 + 200,000 + 100,000 + 0)}{6} = \$250,000$$

$$\text{AAR} = \frac{\$50,000}{\$250,000} = 20\%$$

*Corporate tax rate = t_c . The tax rebate in year 5 of $-\$16,667$ occurs if the rest of the firm is profitable. Here, the loss in the project reduces the taxes of the entire firm.

Example: Accounting Rate of Return

- **STEP 1:** Determining average net income

Average Net Income:

$$(\$100,000 + 150,000 + 50,000 + 0 - 50,000) / 5 = \$50,000$$

- **STEP 2:** Determining average investment

Average Investment:

$$(\$500,000 + 400,000 + 300,000 + 200,000 + 100,000 + 0) / 6 = \$250,000$$

- **STEP 3:** Determining AAR

$$\text{AAR} = \$50,000 / \$250,000 = 20\%$$

Accounting Rate of Return

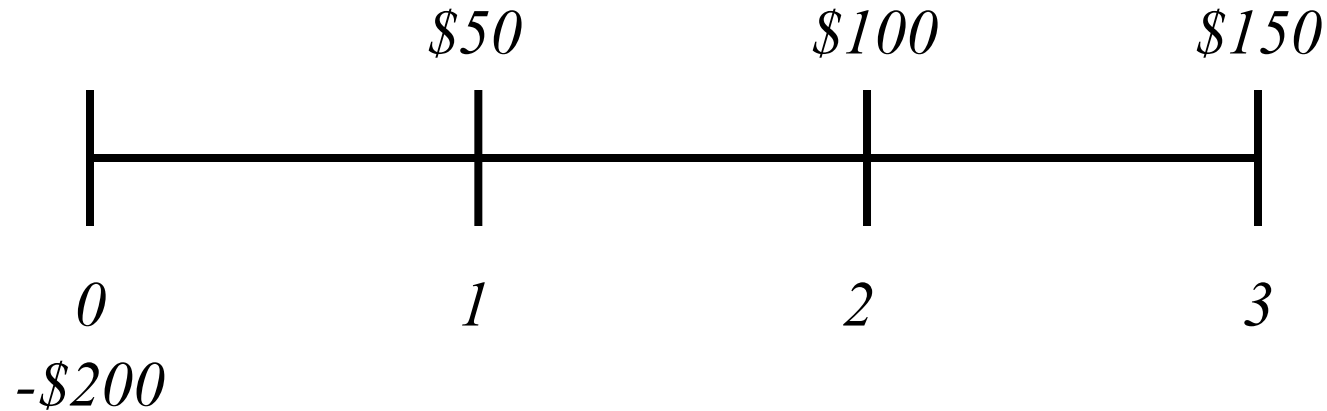
- Disadvantages:
 - It uses accounting profits, not cash flows
 - It ignores the time value of money
 - It does not take risk into account
 - The threshold ARR for accepting or rejecting an investment proposal is arbitrary (Why 20%, not 21% or any other threshold ?)
- Advantages:
 - The required information is readily available
 - It is simple to calculate
 - It considers profits over the entire life of a project
 - It is easy to understand

The Internal Rate of Return

- IRR: the discount rate that sets NPV to zero
- Minimum Acceptance Criteria:
 - Accept if the IRR exceeds the required return
- Ranking Criteria:
 - Select alternative with the highest IRR
- Reinvestment assumption:
 - All future cash flows assumed reinvested at the IRR

IRR: Example

Consider the following project:



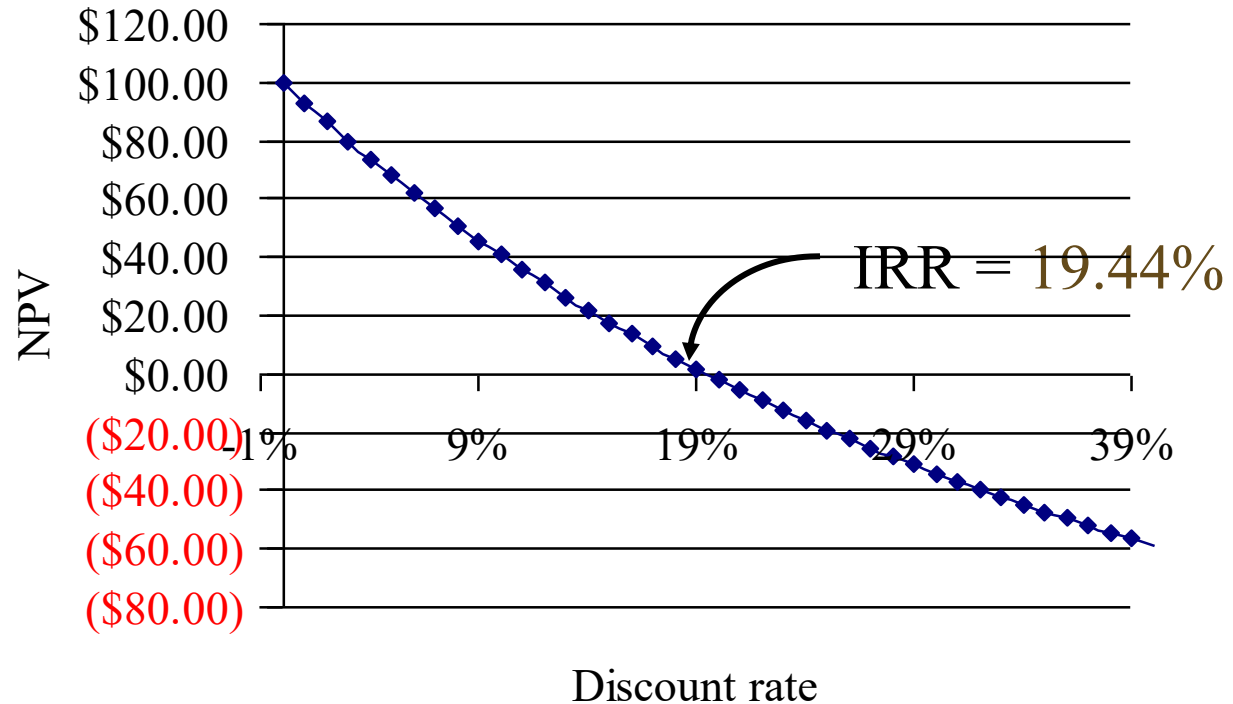
The internal rate of return for this project is 19.44%

$$NPV = 0 = -200 + \frac{\$50}{(1 + IRR)} + \frac{\$100}{(1 + IRR)^2} + \frac{\$150}{(1 + IRR)^3}$$

NPV Payoff Profile

If we graph NPV versus the discount rate, we can see the IRR as the x-axis intercept.

0%	\$100.00
4%	\$73.88
8%	\$51.11
12%	\$31.13
16%	\$13.52
20%	(\$2.08)
24%	(\$15.97)
28%	(\$28.38)
32%	(\$39.51)
36%	(\$49.54)
40%	(\$58.60)
44%	(\$66.82)



Calculating IRR with Spreadsheets

- You start with the cash flows the same as you did for the NPV.
- You use the IRR function:
 - You first enter your range of cash flows, beginning with the initial cash flow.
 - You can enter a guess, but it is not necessary.
 - The default format is a whole percent – you will normally want to increase the decimal places to at least two.



Internal Rate of Return (IRR)

- Disadvantages:
 - Does not distinguish between investing and financing
 - IRR may not exist, or there may be multiple IRRs
 - Problems with mutually exclusive investments
- Advantages:
 - Easy to understand and communicate

Problems with IRR:

Mutually Exclusive vs. Independent

- Mutually Exclusive Projects: only ONE of several potential projects can be chosen, e.g., acquiring an accounting system.
 - RANK all alternatives, and select the best one.
- Independent Projects: accepting or rejecting one project does not affect the decision of the other projects.
 - Must exceed a MINIMUM acceptance criteria

Problems with IRR

- ❑ Investing or Financing

- ❑ Multiple IRRs



All projects

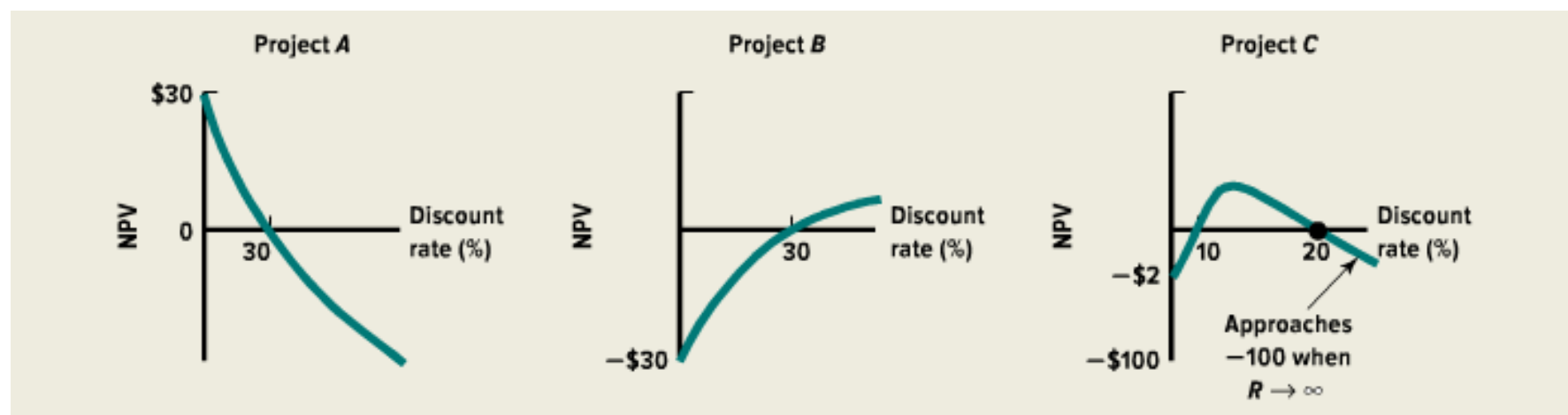
- ❑ The Scale Problem

- ❑ The Timing Problem



Exclusive projects

DATES:	PROJECT A			PROJECT B			PROJECT C		
	0	1	2	0	1	2	0	1	2
Cash flows	-\$100	\$130		\$100	-\$130		-\$100	\$230	-\$132
IRR		30%			30%		10%	and	20%
NPV @10%		\$18.2			-\$18.2			\$0	
Accept if market rate		<30%			>30%		>10%	but	<20%
Financing or investing		Investing			Financing			Mixture	



Investing or financing (Project A and B)

- **The rule of investing projects:**

Accept the project when IRR is **greater** than the discount rate. Reject the project when IRR **is less than** the discount rate .

- **The rule of financing projects:**

Accept the project when IRR is **less** than the discount rate. Reject the project when IRR is **greater** than the discount rate .

Multiple IRRs (Project C)

There are two IRRs for Project C.

Which one should we use?

Modified IRR

- Combining cash flows until only one change in sign remains.
- For Project C, with a discount rate of, say, 14 percent, the value of the last cash flow, -\$132, is $-\$115.79 (-\$132/1.14)$ as of Date 1. Since \$230 is already received at that time, the “adjusted” cash flow at Date 1 is \$114.21 ($-\$230 - 115.79$). Thus the MIRR approach produces the following two cash flows for the project:

(−\$100, \$114.21)

- We are left with only one change in sign. The IRR rule can now be applied. The IRR of these two cash flows is 14.21%, implying that the project should be accepted given our assumed discount rate of 14%.

Problems to mutually exclusive projects-

The Scale Problem

Would you rather make 100% or 50% on your investments?

What if the 100% return is on a \$1 investment, while the 50% return is on a \$1,000 investment?

Example: scale problem

- Stanley Jaffe and Sherry Lansing have just purchased the rights to Corporate Finance: The MotionPicture . They will produce this major motion picture on either a small budget or a big budget. The estimated cash flows are:

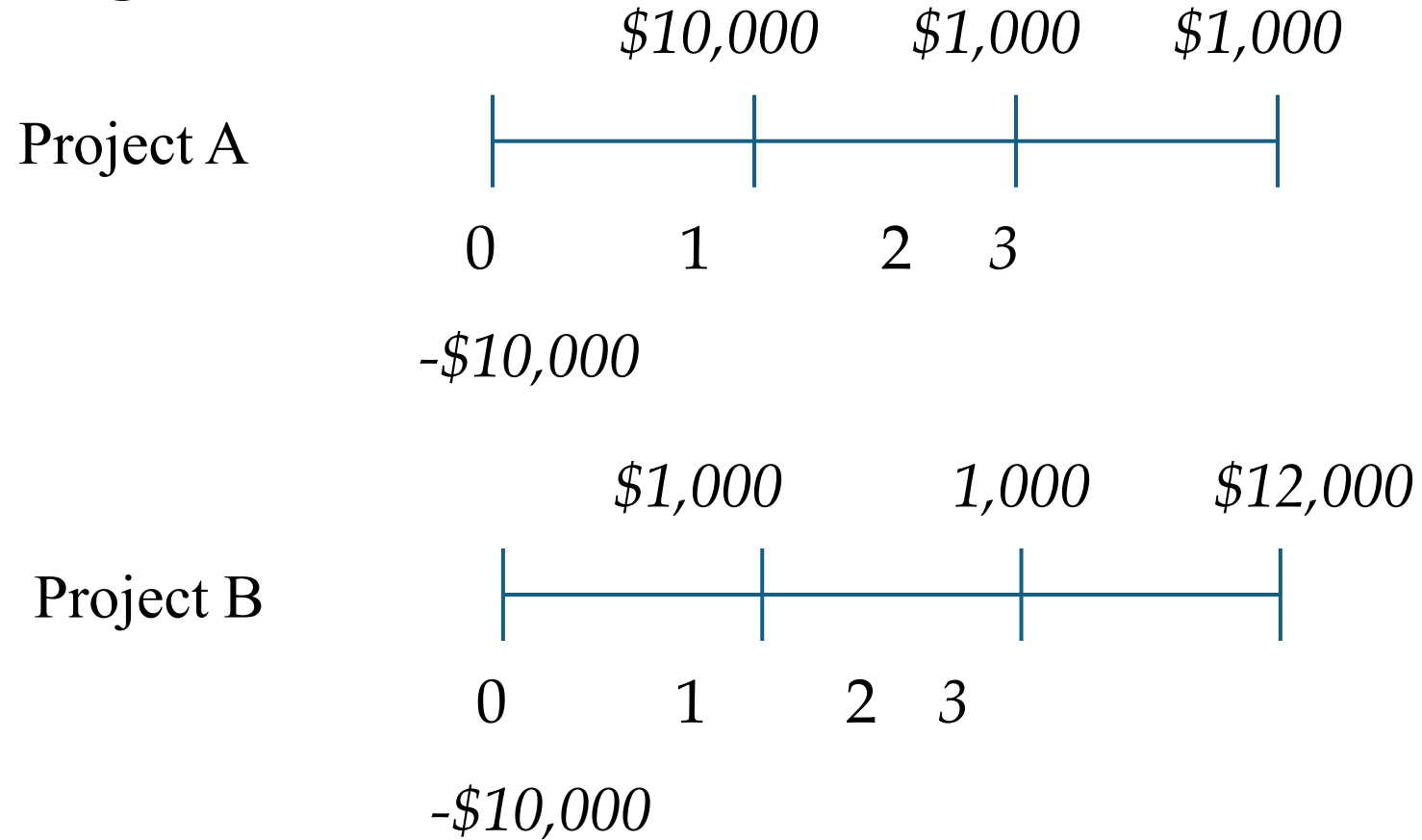
	Cash flow At date 0	cash flow at date 1	NPV @25%	IRR
Small budget	-\$10m	\$40	\$22m	300%
Large budget	-\$25m	65	27m	160

Three ways to solve the scale problems

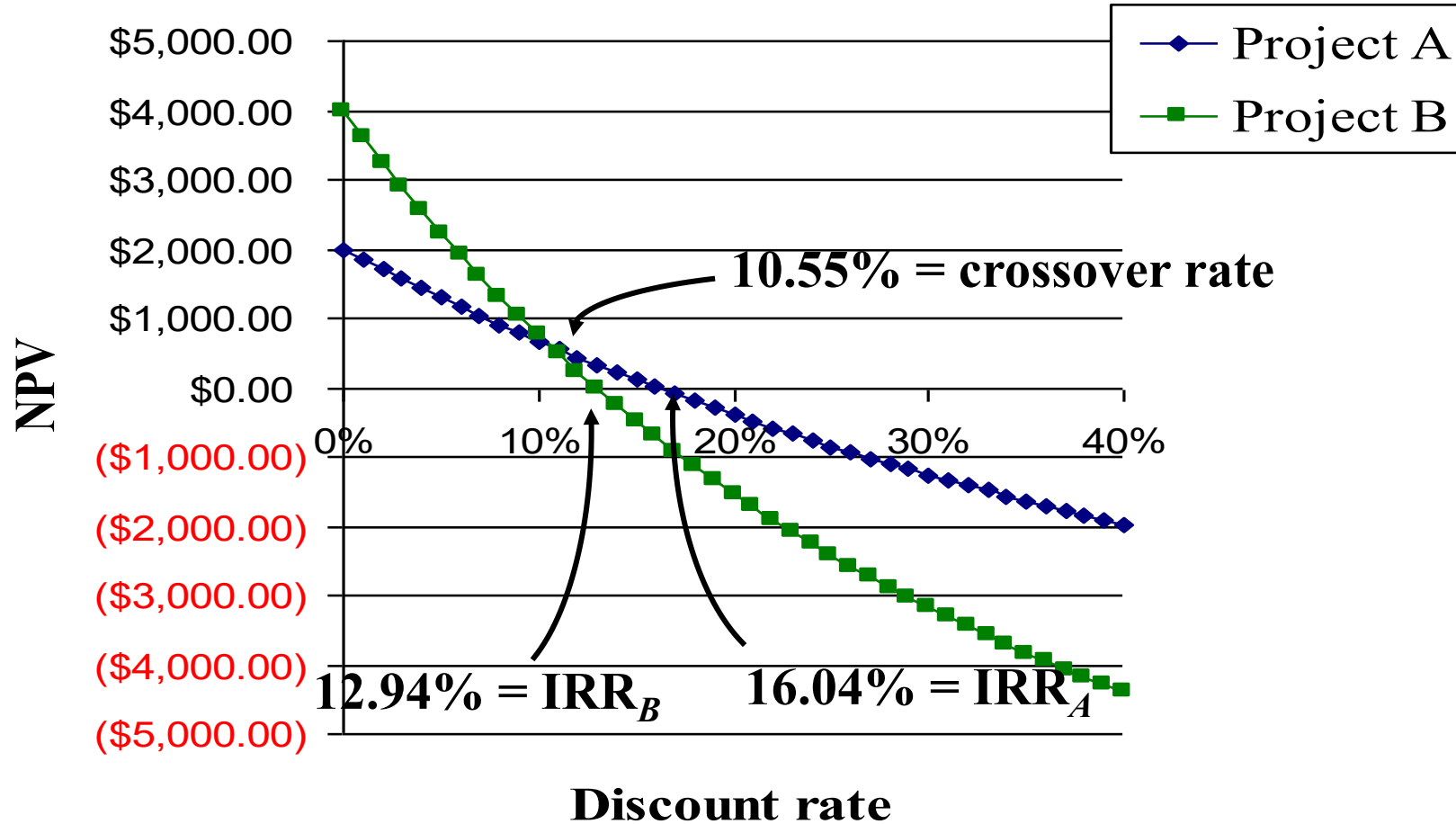
1. Compare the NPVs of the two choices .
2. Calculate the incremental NPV from making the large-budget picture instead of the small-budget picture .
3. Compare the incremental IRR to the discount rate.

Problems to mutually exclusive projects-

The Timing Problem



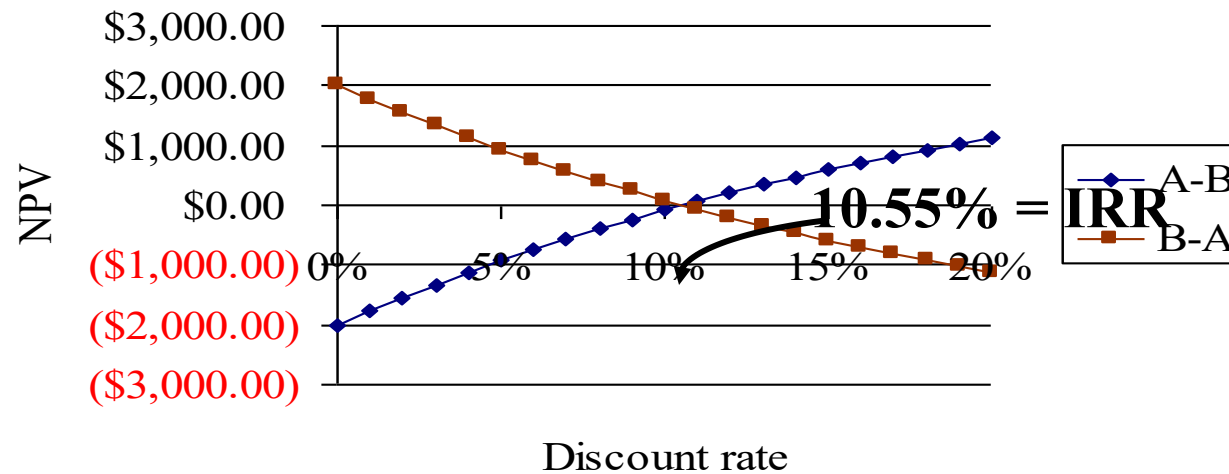
The Timing Problem



Calculating the Crossover Rate

Compute the IRR for either project “A-B” or “B-A”

Year	Project A	Project B	Project A-B	Project B-A
0	(\$10,000)	(\$10,000)	\$0	\$0
1	\$10,000	\$1,000	\$9,000	(\$9,000)
2	\$1,000	\$1,000	\$0	\$0
3	\$1,000	\$12,000	(\$11,000)	\$11,000



NPV versus IRR

- NPV and IRR will generally give the same decision.
- Exceptions:
 - Non-conventional cash flows – cash flow signs change more than once
 - Mutually exclusive projects
 - Initial investments are substantially different
 - Timing of cash flows is substantially different

The Profitability Index (PI)

$$PI = \frac{\text{Total PV of Future Cash Flows}}{\text{Initial Investment}}$$

- Minimum Acceptance Criteria:
 - Accept if $PI > 1$
- Ranking Criteria:
 - Select alternative with highest PI

Capital Rationing: A firm may place a limit on the amount of new investment in a year. The management team has to be rational on how they spend their capital -> Capital Rationing

- If the limit is set by the firm's management or its directors, it is known as **soft capital rationing**
- If the limit is imposed by the financial markets (because investors are not willing to purchase the firm's securities), it is known as **hard capital rationing**
- When a firm is constrained by a dollar limit on new investment, we cannot choose all investments with positive NPV. Instead, we need to select a few that suit our "budget" while still maximizing shareholders' wealth
- If capital rationing exists we evaluate (rank) projects based on the Profitability Index.

Practice

A company has 6 projects but has a budget constraint of \$600,000. Which projects should be accepted?

Project	Initial Cost	NPV
A	200,000	28,000
B	200,000	20,000
C	200,000	15,000
D	200,000	35,000
E	400,000	45,000
F	400,000	22,000
Total	1,600,000	

Example :PI

- Suppose that the Kaufold Corporation has two alternative uses for a warehouse. It can store toxic waste containers (investment *A*) or *electronic equipment (investment B)*. The cash flows are as follows:

	Cash flow			PV	PI	NPV
	C0	C1	C2	@12%		@12%
A	-\$20	\$70	\$10	\$70.5	3.53	\$50.5
B	-\$10	15	40	45.3	4.53	35.3

The Profitability Index

- Disadvantages:
 - Problems with mutually exclusive investments
- Advantages:
 - Be useful when available investment funds are limited (Capital rationing)
 - Easy to understand and communicate
 - Correct decision when evaluating independent projects

The Practice of Capital Budgeting

- Varies by industry:
 - Some firms use payback, others use accounting rate of return.
- The most frequently used technique for large corporations is IRR or NPV.

	% ALWAYS OR ALMOST ALWAYS
Internal rate of return (IRR)	75.6%
Net present value (NPV)	74.9
Payback method	56.7
Accounting rate of return	30.3
Discounted payback	29.5
Profitability index	11.9

Example of Investment Rules

Compute the IRR, NPV, PI, and payback period for the following two projects. Assume the required return is 10%.

<u>Year</u>	<u>Project A</u>	<u>Project B</u>
0	-\$200	-\$150
1	\$200	\$50
2	\$800	\$100
3	-\$800	\$150

Example of Investment Rules

	Project A	Project B
CF_0	-\$200.00	-\$150.00
PV_0 of CF_{1-3}	\$241.92	\$240.80
NPV =	\$41.92	\$90.80
IRR =	0%, 100%	36.19%
PI =	1.2096	1.6053

Example of Investment Rules

Payback Period:

Time	CF	<i>Project A</i>		<i>Project B</i>	
		Cum. CF	CF	Cum. CF	CF
0	-200	-200	-200	-150	-150
1	200	0	0	50	-100
2	800	800	800	100	0
3	-800	0	0	150	150

Payback period for project B = 2 years.

Payback period for project A = 1 or 3 years?

NPV and IRR Relationship

<u>Discount rate</u>	<u>NPV for A</u>	<u>NPV for B</u>
-10%	-87.52	234.77
0%	0.00	150.00
20%	59.26	47.92
40%	59.48	-8.60
60%	42.19	-43.07
80%	20.85	-65.64
100%	0.00	-81.25
120%	-18.93	-92.52

NPV Profiles

